hw1

October 18, 2025

1 ChatGPT questions

Put this question ChatGPT and get the answer. You are allowed to discuss further. Please put a link to the response and your conversation. Given the response, you must first say whether: (i) the response is correct, (ii) the response is false, or (iii) you are really not sure. In the first case, give a succinct explanation of the proof. In the second case, explain why it is false. In the third case, explain your confusion. You do not need to write too much.

1. Give a bijection from $\{x|x > 1, x \in \mathbb{R}\}$ and the interval $(0, 1)$.

   claude 3.5 response

   I'll solve this by constructing a bijective function between the set of real numbers greater than 1 and the open interval (0,1).

   (a) Bijection Construction

   A classic and elegant solution to this problem is to use the following function:
   $f(x) = \frac{1}{x-1}$

   (b) Proof of Bijectivity

   i. Injective (One-to-One) Let $x_1, x_2 \in \{x|x > 1, x \in \mathbb{R}\}$ If $f(x_1) = f(x_2)$, then: $\frac{1}{x_1-1} = \frac{1}{x_2-1}$ This implies $x_1 = x_2$, proving injectivity.

   ii. Surjective (Onto) For any $y \in (0, 1)$, we can find an x such that $f(x) = y$: $y = \frac{1}{x-1}$ $x = \frac{1}{y} + 1$

   Note that: - When $y \in (0, 1)$, $x = \frac{1}{y} + 1 > 1$ - $f(x) = \frac{1}{(\frac{1}{y}+1)-1} = y$

   Thus, f maps $\{x|x > 1, x \in \mathbb{R}\}$ bijectively onto $(0, 1)$.

   End response

   the response is correct. it shows that for $x > 1, x \in \mathbb{R}$, the map $f(x) = \frac{1}{x-1}$ is injective and surjective to the interval $(0, 1)$.

1. NAE-3SAT is the language of 3-CNFs with an assignment that does not make all literals in a clause have the same value. Prove that NAE-3SAT is $\mathbb{NP}$-complete.

2. Prove that if FACTORING is co-$\mathbb{NP}$-complete, then $TAUT \leq_{p(n)} SAT$ (recall that $TAUT$ is the language of tautologically true Boolean formulas.)